

NetFRAME Infrastructure Update

June 21, 2026

Randy Commissioning · GPU Swap · DUNE Agent Planning

Randy — SuperMicro SYS-2028U-E1CNRT+

Hardware Inventory

Component	Detail
Motherboard	Supermicro X10DRU-i+
Chassis	SYS-2028U-E1CNRT+ (219U-10)
Board Serial	OM15CS036459
CPUs	Dual Intel Xeon E5-2690 v4 @ 2.60 GHz
RAM	128GB ECC DDR4 2133 MHz
BIOS	3.5 (flashed from 2.0 on 06/21/2026)
IPMI	192.168.10.22, BMC firmware 03.94, user ADMIN
NIC	Mellanox ConnectX-3 EN 10GbE (MCX312A-XCB7)
RAID Card	LSI MegaRAID (MR RMB L3-25376-00A) + Tecate supercapacitor
GPU	AMD RX 580 8GB DDR5 (installed, awaiting power cable)
GPU PWR Headers	3× onboard (GPU PWR2/3/4) — SuperMicro proprietary
Drive Bays	2.5" native bays, 22 caddies total
Riser	RSC-R10W-E8R

BIOS Flash — Completed

Method: UEFI Shell via IKVM console (IPMI web UI at 192.168.10.22)

1. Downloaded X10DRU2.427.zip from Supermicro support portal
2. Created FAT32 UEFI bootable USB on Ares (/dev/sda) via parted + mkfs.fat
3. Copied UEFI folder contents to /EFI/BOOT/ on USB; added BOOTX64.EFI
4. Fixed broken Java KVM via IKVM Reset (Maintenance → IKVM Reset in IPMI)
5. Booted Randy to UEFI Built-in EFI Shell via F11 boot menu
6. Navigated: fs0: → cd EFI\BOOT → flash.nsh X10DRU2.427
7. Two-stage flash: FDT difference detected → reboot → STARTUP.NSH auto-resumed
8. **Result:** BIOS 3.5, ME Entire Image update success

Key lesson: Supermicro BIOS jumps with different FDT require two boot cycles. First run creates STARTUP.NSH and reboots. Boot back to EFI Shell and let STARTUP.NSH run automatically without pressing ESC.

Parts Ordered — June 21, 2026

Part	Part Number	Purpose	Qty
Dell GPU power cable	N08NH	Aux power for QuarkyLab dual RTX 6000	2
SuperMicro GPU cable	CBL-PWEX-0665	8-pin CPU → 8-pin PCIe for RX 580 in Randy	1
SuperMicro drive caddy	MCP-220-00075-0B	2.5" native caddies for Randy	2

Randy's Role in Cluster

Randy is the **infrastructure backbone** — not a compute node.

Services (priority order):

1. **Proxmox Backup Server (PBS)** — critical; cluster currently has no backup
2. **NFS/ZFS storage** — serves DS4246 JBOD + internal bays to cluster via NFS
3. **Jellyfin** — media server; RX 580 handles transcoding via AMD ROCm
4. **Prometheus + Grafana + Loki** — migrate from management G4 EliteDesk
5. **Scrutiny** — drive health monitoring for 22+ drives
6. **MinIO** — S3-compatible object storage for Fernanda's datasets
7. **Vaultwarden** — password manager (current node TBD)

Randy does NOT run: ML/LLM workloads, Ollama inference, or GPU compute. The RX 580 is for display output and media transcoding only.

Pending Before Proxmox Install

CBL-PWEX-0665 arrives → power RX 580

Confirm video output via RX 580 HDMI → KVM drawer

Install Proxmox via external USB (not internal thumb drive)

Determine LSI MegaRAID fate (flash to IT mode vs retain for RAID)

Add Randy to Proxmox cluster

GPU Swap Decision**New GPU Configuration**

Server	GPU	VRAM	CUDA	Role
QuarkyLab	RTX 8000 (from Jarvis)	48GB	Yes	Fernanda DUNE agent
Jarvis	2× RTX 6000 (from QuarkyLab)	48GB total	Yes	LLM inference/Ollama
Randy	RX 580	8GB	No (ROCm)	Display + transcoding

Rationale

QuarkyLab gets RTX 8000: Fernanda's DUNE RAG agent needs maximum single-GPU VRAM. 48GB holds a large code model and embedding model simultaneously without splitting. Single GPU means cleaner inference with no PCIe overhead.

Jarvis gets dual RTX 6000: LLM inference workloads are bursty, not sustained. Ollama handles multi-GPU natively. 48GB aggregate VRAM matches the RTX 8000. PCIe overhead is acceptable for the `llm_router` query routing use case.

NVLink: Not required for either server. Fernanda's workload is inference and RAG retrieval, not distributed training. No unified VRAM pool is needed.

Fernanda's DUNE Agent Project

Overview

- **Goal:** RAG agent over the DUNE experiment codebase
- **Purpose:** Help new scientists understand the full codebase during onboarding at SURF (South Dakota)
- **Host:** QuarkyLab (after RTX 8000 swap)
- **Claude Project:** "DUNE Agent — Fernanda" (created 06/21/2026)

Architecture

- RAG pipeline (inference workload, not training)
- Embedding model dedicated to RTX 8000 GPU allocation
- Inference model on RTX 8000 (48GB accommodates large code model)
- Suggested models: Qwen2.5-Coder 32B unquantized or 72B Q4_K_M
- Vector store: ChromaDB or Qdrant (TBD)

Sandbox / Lab Node

- **Hardware:** Spare HP EliteDesk G4 (unused, in rack)
- **Status:** Standalone — do NOT add to Proxmox cluster
- **Purpose:** Safe learning environment; CCNA lab; service testing before Randy deployment
- **Tools:** GNS3, EVE-NG, k3s (future)
- **Network:** VLAN 70 (lab), IP range 192.168.70.x
- **Hostname:** TBD

Claude Projects Created

Project	Scope
NetFRAME — LLM Infrastructure	Jarvis, Ollama, <code>llm_router</code> , GPU swap
DUNE Agent — Fernanda	RAG pipeline, codebase ingestion, QuarkyLab ML env

Full Cluster GPU Inventory

Server	GPU	VRAM	CUDA	ROCm	Primary Use
QuarkyLab	RTX 8000	48GB	Yes	No	Fernanda ML / DUNE agent
Jarvis	2× RTX 6000	48GB total	Yes	No	LLM inference (Ollama)
Randy	RX 580	8GB	No	Yes	Display + Jellyfin transcode

Every node in the cluster now has a GPU. This enables distributed inference routing across all GPU nodes via `llm_router.py` and provides full cluster-wide redundancy for GPU workloads.